

HW9 - Computer Simulation II

Schewtschik, Guilherme (3748052)

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1

Using the following function to reweigh the observables:

```
def reweight_moments(energies, magnetizations, L,
                    beta0, beta_values):
    V = L * L

    m = magnetizations / V

    absm = np.abs(m)

    m2 = m ** 2
    m4 = m ** 4

    results = []

    for beta in beta_values:

        delta = beta - beta0

        # logarithmic weights improve numerical stability
        logw = -delta * energies
        logw -= np.max(logw)

        w = np.exp(logw)

        Z = np.sum(w)

        mean_absm = np.sum(w * absm) / Z
        mean_m2 = np.sum(w * m2) / Z
        mean_m4 = np.sum(w * m4) / Z

        results.append(
            {
                "beta": beta,
                "absm": mean_absm,
                "m2": mean_m2,
                "m4": mean_m4,
            }
        )

    return pd.DataFrame(results)
```

We obtain the following graph:

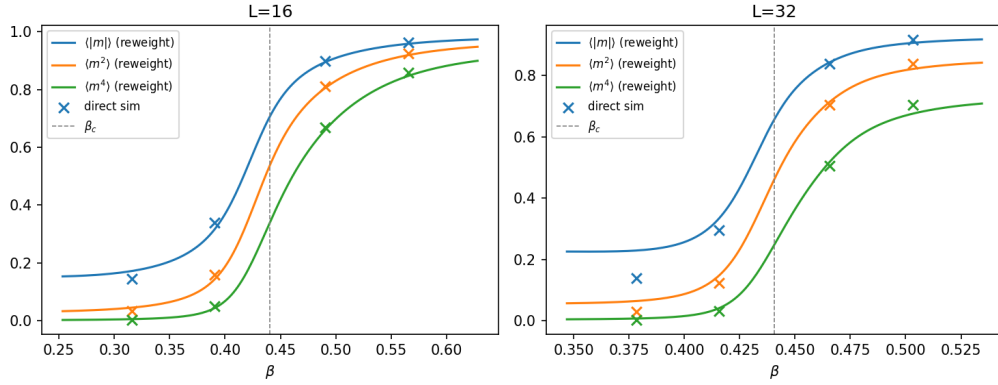


Figure 1: Magnetizations.

We can see the reweighted observables stay close to the actual simulation and display the expected behaviour becoming small below β_c and approaching 1 above the critical temperature, we also see that the higher powers of the magnetization stay closer to zero in the unorderd phase and need a higher temperature to truly magnetize. The size scaling shows that with bigger lattice sizes this effect is even more exaggerated, where the different powers of the magnetization stay separate for longer in the $L = 32$ case.

2

Simulating the potts model for $L = 16, 32, 64, 128$, and using the binning method to estimate the auto correlation time

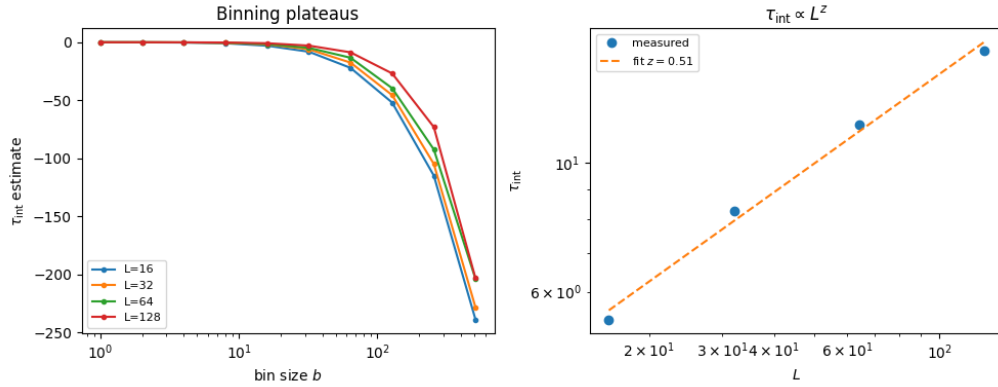


Figure 2: Autocorrelation time.

The binning plateaus confirm that τ_{int} is small for all system sizes. A log-log fit of τ_{int} and L yields the value for $z \approx 0.51$. The fitted z value is noticeably larger than the value observed for the Wolff algorithm on the 2D Ising model of $z \approx 0.25$, which is expected.